Question	Answer	Marks	Partial Marks
1(a)	Rotation 90° clockwise oe [centre] (5, 2)	**3**	**B1** for each
1(b)(i)	Translation $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$	**2**	**B1** for each
1(b)(ii)	4.12 or 4.123...	**2**	**M1** for (*their* (–1))^2 + (*their* 4)^2
2(a)	52°	**3**	**M1** for 180 – 2 × 38, implied by 104 **M1** for *their AOB* ÷ 2
2(b)(i)	80°	**2**	**B1** for *FEC* =50 or *FCE* = 50
2(b)(ii)	100°	**1**	**FT** 180 – *their* **(i)**
3(a)(i)	4.095	**2**	**B1** for figs 4095 or **M1** for $\frac{525 \times 7.8}{1000}$
3(a)(ii)	15	**3**	**B2** for 35 OR **M2** for $\frac{1}{2}(10+4)\times 5 \times L = 525$ oe **M1** for $\frac{1}{2}(10+4)\times 5$ oe
3(a)(iii)	455 or 454.9...	**6**	**M3** for *their* [*BD* =] $\sqrt{3^2+5^2}$ × (*their* 15) [× 2] or **B2** for $\sqrt{34}$ or 5.83 or 5.830 to 5.831 or **M1** for $5^2 + \left(\frac{1}{2}(10-4)\right)^2$ and **M1** for *their* 35 × 2 **M1** for (*their* 15) × 10 and (*their* 15) × 4
3(a)(iv)	4200	**3**	**M2** for $525 \times \left(\frac{10}{5}\right)^3$ oe or **M1** for $\left(\frac{10}{5}\right)^3$ or $\left(\frac{5}{10}\right)^3$ oe

Question	Answer	Marks	Partial Marks
3(b)	182.875 ... 307.125 final answer	**3**	**B2** for either seen or **M1** for 10 ± 0.5 or 6 ± 0.5 or 4 ± 0.5 oe
4(a)	Correctly eliminate one variable	**M1**	
	$p = 3$ $q = -1$	**A2**	**A1** for each If M0, **SC1** for 2 values satisfying one of original equations If 0 scored **SC1** for correct answers with no working
4(b)	$1\frac{1}{11}$ or $\frac{12}{11}$ 1.09 or 1.090 to 1.091	**2**	**M1** for $\frac{3x}{12}+\frac{8x}{12}=1$ or better
4(c)(i)	$-2 < x \leqslant 3$	**3**	**B2** for $-2 < x$ or $x \leqslant 3$ or **M1** for $-8 + 2 < 3x$ or $3x \leqslant 7 + 2$
4(c)(ii)	–1, 0, 1, 2, 3	**1**	**FT** dep on –ve and +ve values in *their* **(c)(i)**
4(d)	$4a(4-a)$ final answer	**2**	**B1** for any correct partial factorisation
4(e)(i)	$\frac{2b}{3a}$ final answer	**2**	**M1** for $\frac{1}{2a}\times\frac{4b}{3}$ or better
4(e)(ii)	$\frac{x-2}{x-1}$ final answer nfww	**2**	**B1** for $2(x-1)-x$ oe seen.
5(a)(i)	105	**2**	**M1** for $\frac{3}{100}\times 500[\times 7]$
5(a)(ii)	115 or 114.9...	**3**	**M2** for $500\times\left(1+\frac{3}{100}\right)^7[-500]$ or **M1** for $500\times\left(1+\frac{3}{100}\right)^k$, k integer $\geqslant 2$
5(b)	8600	**3**	**M2** for $\frac{6269.4}{\left(1-\frac{10}{100}\right)^3}$ oe or **M1** for $C\times\left(1-\frac{10}{100}\right)^3=6269.4$ oe
6(a)	9.33 or 9.334...	**3**	**M2** for $\frac{12\sin 50}{\sin 100}$ or **M1** for $\frac{\sin 100}{12}=\frac{\sin 50}{AD}$ oe